KAIZEN MATHS

Lesson starter

Roots of Equations and Transformations of Roots

Quadratic Roots

Name

S1

Use the relationships between coefficients and roots.

The roots of $2x^2 + 10x + 16 = 0$ are α and β. Find $\alpha + \beta$ and $\alpha\beta$.

S2

Form each quadratic equation from the given roots.

Form a quadratic equation with roots -7 and -3.

S3

Use the given root information to find the unknown coefficients.

The equation $2x^2 + px + q = 0$ has roots α and β with $\alpha + \beta = 3$ and $\alpha\beta = -10$. Find p and q.

S4

Use $\alpha + \beta$ and $\alpha\beta$ to evaluate each expression.

The roots of $x^2 - 6x + 8 = 0$ are α and β. Find $\alpha^2 + \beta^2$.

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Roots of Equations and Transformations of Roots

Quadratic Roots

Name _____

EA1

Use the relationships between coefficients and roots.

The roots of $3x^2 + 30 = 0$ are α and β. Find $\alpha + \beta$ and $\alpha\beta$.

EA2

Form each quadratic equation from the given roots.

Form a quadratic equation with roots 4 and 3.

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Roots of Equations and Transformations of Roots

Quadratic Roots

Name _____

EB1

Use the given root information to find the unknown coefficients.

The equation $4x^2 + px + q = 0$ has roots α and β with $\alpha + \beta = -1$ and $\alpha\beta = -6$. Find p and q.

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Roots of Equations and Transformations of Roots

Quadratic Roots

Name _____

EA1

Use the relationships between coefficients and roots.

The roots of $3x^2 + 30 = 0$ are α and β. Find $\alpha + \beta$ and $\alpha\beta$.

EA2

Form each quadratic equation from the given roots.

Form a quadratic equation with roots 4 and 3.

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Roots of Equations and Transformations of Roots

Quadratic Roots

Name _____

EB1

Use the given root information to find the unknown coefficients.

The equation $4x^2 + px + q = 0$ has roots α and β with $\alpha + \beta = -1$ and $\alpha\beta = -6$. Find p and q.

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Teacher answers & working

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S1

Use the relationships between coefficients and roots.

The roots of $2x^2 + 10x + 16 = 0$ are α and β. Find α + β and αβ.

Answer:

$\alpha + \beta = -5, \alpha\beta = 8$

Working:

For $ax^2 + bx + c = 0$, $\alpha + \beta = \frac{-b}{a}$ and $\alpha\beta = \frac{c}{a}$.

Here a = 2, b = 10, c = 16.

$\alpha + \beta = \frac{-(10)}{2} = -5.$

$\alpha\beta = \frac{16}{2} = 8.$

S2

Form each quadratic equation from the given roots.

Form a quadratic equation with roots -7 and -3.

Answer:

$x^2 + 10x + 21 = 0$

Working:

If the roots are α and β, then the equation is $x^2 - (\alpha + \beta)x + \alpha\beta = 0$.

$\alpha + \beta = -7 + -3 = -10.$

$\alpha\beta = -7 \times -3 = 21.$

So the equation is $x^2 - (-10)x + (21) = 0$.

Therefore $x^2 + 10x + 21 = 0$.

S3

Use the given root information to find the unknown coefficients.

The equation $2x^2 + px + q = 0$ has roots α and β with α + β = 3 and αβ = -10. Find p and q.

Answer:

$p = -6, q = -20$

Working:

Compare with $ax^2 + bx + c = 0$.

Here a = 2, b = p and c = q.

$\alpha + \beta = \frac{-p}{2} = 3.$

So −p = 6, giving p = -6.

$\alpha\beta = \frac{q}{2} = -10.$

So $q = 2 \times -10 = -20.$

Teacher answers & working

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S4

Use α + β and αβ to evaluate each expression.

The roots of $x^2 - 6x + 8 = 0$ are α and β. Find $\alpha^2 + \beta^2$.

Answer:

20

Working:

First use the root relationships.

$\alpha + \beta = 6$ and $\alpha\beta = 8$.

Use $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$.

$\alpha^2 + \beta^2 = (6)^2 - 2(8)$.

$= 36 - 16 = 20$.

EA1

Use the relationships between coefficients and roots.

The roots of $3x^2 + 30 = 0$ are α and β. Find α + β and αβ.

Answer:

$\alpha + \beta = 0, \alpha\beta = 10$

Working:

For $ax^2 + bx + c = 0$, $\alpha + \beta = \frac{-b}{a}$ and $\alpha\beta = \frac{c}{a}$.

Here a = 3, b = 0, c = 30.

$\alpha + \beta = \frac{-(0)}{3} = 0$.

$\alpha\beta = \frac{30}{3} = 10$.

EA2

Form each quadratic equation from the given roots.

Form a quadratic equation with roots 4 and 3.

Answer:

$x^2 - 7x + 12 = 0$

Working:

If the roots are α and β, then the equation is $x^2 - (\alpha + \beta)x + \alpha\beta = 0$.

$\alpha + \beta = 4 + 3 = 7$.

$\alpha\beta = 4 \times 3 = 12$.

So the equation is $x^2 - (7)x + (12) = 0$.

Therefore $x^2 - 7x + 12 = 0$.

Teacher answers & working

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EB1

Use the given root information to find the unknown coefficients.

The equation $4x^2 + px + q = 0$ has roots α and β with $\alpha + \beta = -1$ and $\alpha\beta = -6$. Find p and q.

Answer:

$p = 4$, $q = -24$

Working:

Compare with $ax^2 + bx + c = 0$.

Here a = 4, b = p and c = q.

$\alpha + \beta = \frac{-p}{4} = -1$.

So −p = -4, giving p = 4.

$\alpha\beta = \frac{q}{4} = -6$.

So $q = 4 \times -6 = -24$.